Hypothesis Testing – One-Way ANOVA (Regional Sales Analysis)

Objective

To determine whether there is a statistically significant difference in sales among the four regions: West, East, Central, and South.

Hypotheses

Null Hypothesis (H_0):

There is no significant difference in average sales across regions.

Alternative Hypothesis (H_1):

There is a significant difference in average sales across regions.

Significance Level

$\alpha = 0.05$

Test Used

One-Way ANOVA (Single Factor)

Anova: Single Factor

SUMMARY

Groups	Count	Sum	Average	Variance
Sales - We	3140	710219.7	226.1846	274828.4
Sales - Eas	2785	669518.7	240.4017	392334.5
Sales - Cer	2277	492646.9	216.3579	404547.1
Sales - Sou	1598	389151.5	243.5241	608166.9

ANOVA

Source of Varia	SS	df	MS	F	P-value	F crit
Between G	1057278	3	352426	0.897432	0.441578	2.605816
Within Grc	3.85E+09	9796	392704.9			
Total	3.85E+09	9799				

Reason:

ANOVA is used because we are comparing the means of more than two independent groups (four regions).

ANOVA Results

F-value = 0.8974

P-value = 0.4416

F-critical = 2.6058

Decision Rule

If P-value < 0.05

➡Reject H_o

➡Means regional sales difference significant

If P-value > 0.05

➡Fail to reject H_o

➡Means difference not statistically significant

Since 0.4416 > 0.05, we fail to reject the null hypothesis.

Compare F and F Critical

Rule:

- If **F > F Critical**, reject the null hypothesis (H_o)
- If **F < F Critical**, fail to reject (accept) the null hypothesis (H_o)

In this case:

- **F = 0.8974**
- **F Critical = 2.6058**

Since **0.8974 < 2.6058**, we **fail to reject the null hypothesis (H_0)**.

Conclusion

The analysis shows that there is no statistically significant difference in sales among the four regions. Regional location does not appear to significantly influence sales performance.

Business Interpretation

Although average sales vary slightly between regions, the variation is not statistically significant. Therefore, business strategies should focus more on product categories, promotions, and seasonal trends rather than geographical differences.